THE HONG KONG UNIVERSITY OF SCIENCE AND TECHNOLOGY
Department of Electronic and Computer Engineering
ELEC 1100

Laboratory 6: Assembly of the Robot Car (4%)

A) Objectives:

- By applying all techniques learned in the previous labs, design and complete the control circuit of the robot vehicle so that it can track along a racing track.

B) Hardware and Software:

- Robot Car Kit with tools (with line sensors)
- Nano-Board (Arduino compatible) and open-source **Arduino Software** (**IDE 1.8.19**)

C) Experimental Procedures:

Experiment 1: Build the Robot Car (~40min)

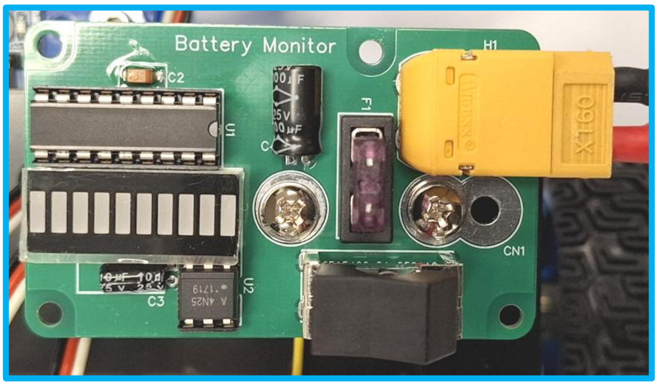
Step 1: Download the **"Robot Car Guide_Full Assembly"** file from your Canvas lab page. A demonstration video for the full assembly is also available on Canvas for your reference.

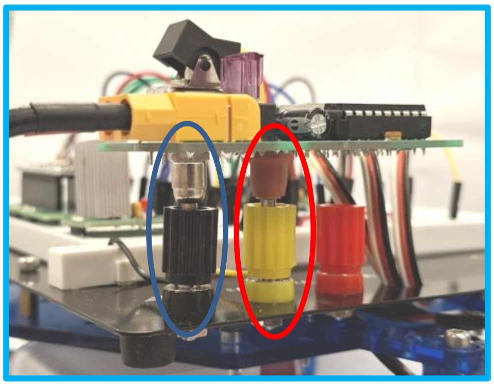
Step 2: Assemble the full version of your Robot Car following the guide or demo video.

****** TA Check 1: Show the Robot Car to your TA ******

Experiment 2: Confirm the Voltage on the Breadboard (~20 mins)

Step 1: Connect the **~12V battery** to the battery monitor. Insert the battery monitor into the breadboard's power socket, then connect its **positive** output to the **12V bus** and its **negative** output to the **GND bus**, as you set up in Lab 4 or Lab 5.





Notice

- This 12V battery is the power source for the complete circuit on your breadboard.
- Anywhere a 5V is needed it should be connected to the **LM7805** output through your breadboard 5V-bus.

Step 2: Press the **black button** on the battery monitor to turn it on. Then measure the voltage at each pin listed in the table below to verify your circuit connections.

Note: During the project period, you will need to refer to the project "**Supplementary**" file to charge the battery regularly and properly to power your robot car system.

However, do **NOT charge your car battery in Lab 6 class.**

It takes more than 8 hours to fully charge a low energy level battery.

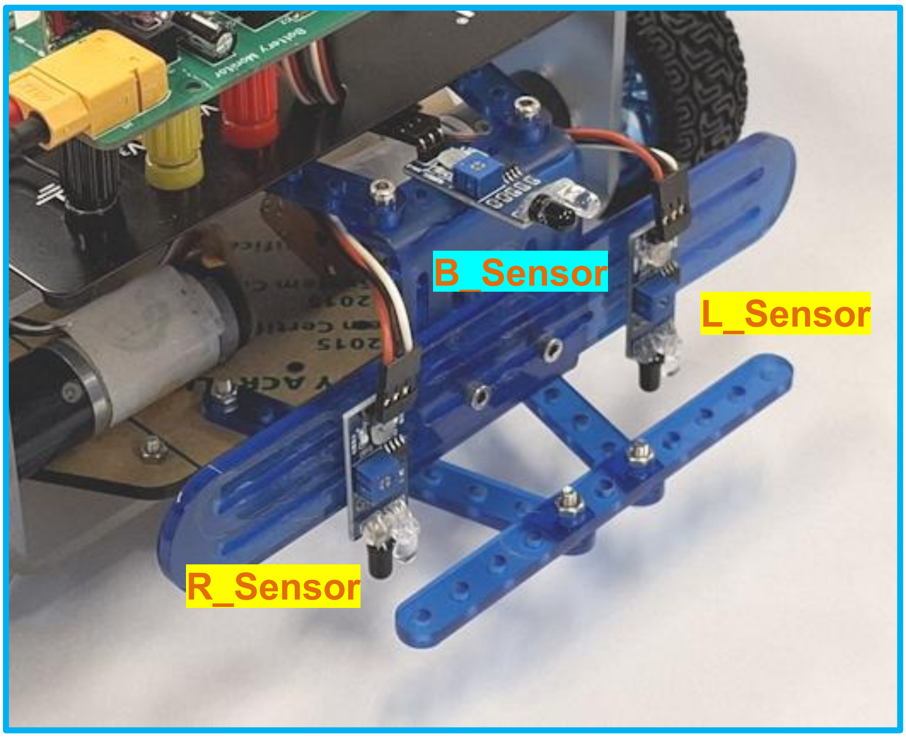
For Lab 6, you may use the DC power supply to provide 12V if the car battery is low.

****** TA Check 2: Demo to your TA that the voltage measurements on your breadboard match with the table below to confirm your Lab 5 circuit connections ******

LM7805 Regulator		**L293** Motor Driver		**74HC14** Inverter		**Nano-board**
V_{in}	V_{out}	Pin **8**	Pin **16**	Pin **7**	Pin **14**	**+5V** terminal
12V	**5V**	**12V**	**5V**	**0V**	**5V**	**5V**

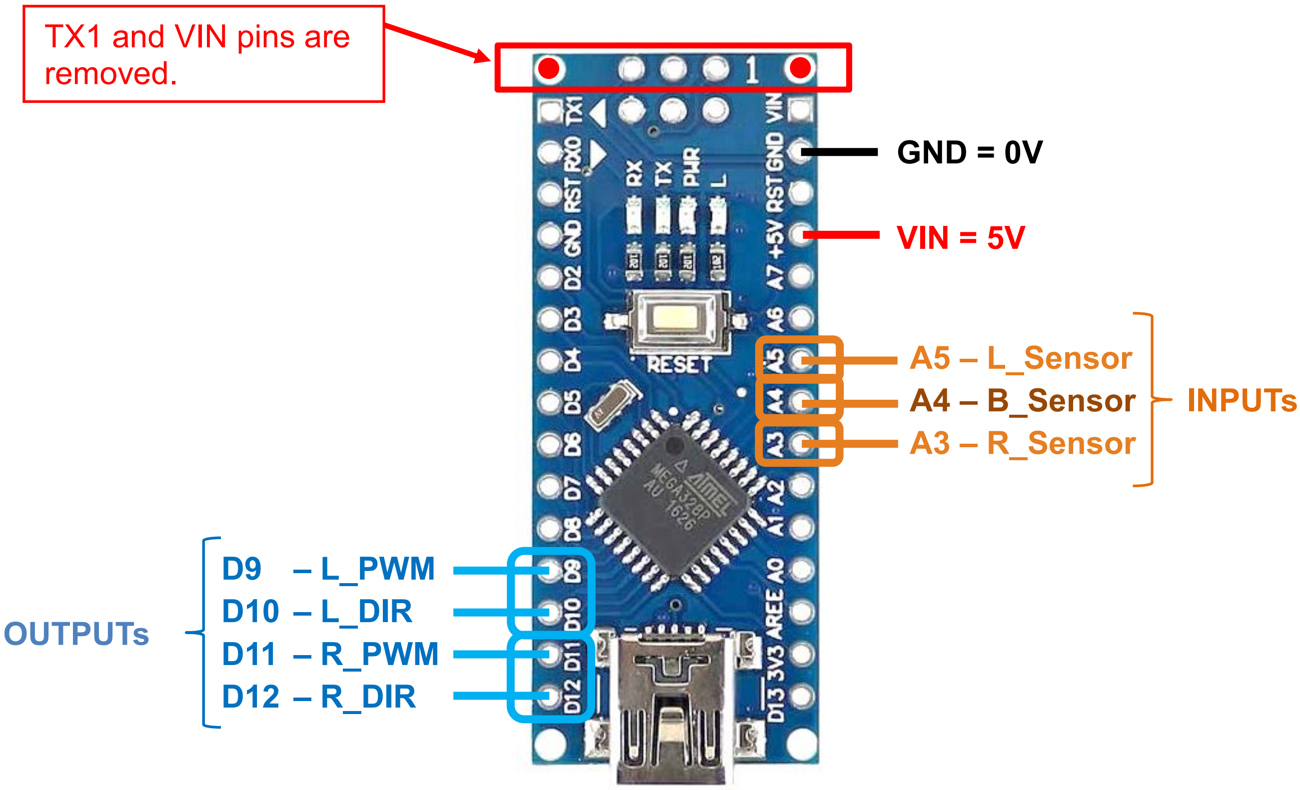
Experiment 3: The measurements of Car Sensors (~20 min)

Step 1: Install the three sensors on your car as shown in the demo video and the photo below. One sensor will serve as the **bumper sensor (B_Sensor)** to start the robot car. The other two sensors—the **left (L_Sensor)** and **right (R_Sensor)**—should face downward to detect the racing track.



Step 2: Power and ground each sensor to the **5V-bus** or **GND-bus** on your breadboard and connect sensor outputs (from sensor "OUT" pin) to the **Nano-board** as shown below

You may refer to the **example breadboard layout** provided in the **tutorial notes**.



Step 3: **Disconnect the two motors from your breadboard for now.**

Step 4: Examine your car sensors one by one:

- Bumper sensor: put your hand (or a white paper) in front of the sensor, the external LED is "ON" when seeing "white" and could be turned "OFF" when not seeing "white".

- Left and right sensors: move your car across a white line on the black mat, the external LED is "ON" when seeing "white" and "OFF" when seeing "black".

  If not, do the followings:

  - modify the **sensitivity of the sensor** by adjusting the variable resistor using a screwdriver.
  - adjust the **sensor height** (the height of your Car Front Bar) from the racing track.

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TA Check 3: Show demo to your TA that:

(1) The bumper sensor can respond to when your hand (or white paper) is in front of it.

(2) Each of the left & right sensors can differentiate the white line from the black mat.

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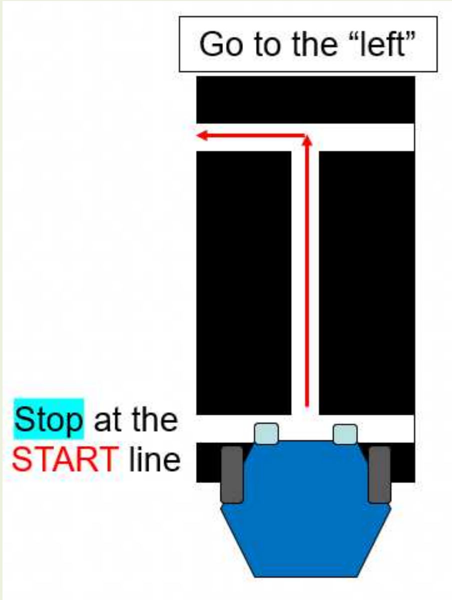
Experiment 4: Robot Car Motion Control (~20 min)

In this experiment, you are going to program the Nano-board to control your robot car to move along a racing track.

The racing track is a white line on a black mat as shown below.

➢ **Your tasks are to let the robot car achieve:**

Task No.	Tasks
1	After power on, car wheels should be **STOP** after board initialization.
2	Put the car at the start position with all front-bar sensors ON the horizontal START white line. The car should still be at **STOP** state.
3	The **bumper sensor** is triggered to start the robot car. Follow the straight line until reaching the 'T' junction.
4	Navigate the "go left" turning at the junction.



Before the coding work, you are required to power OFF your circuit and double-check your breadboard connection.

Step 1: Confirm that the **B_Sensor** (sensor "OUT" pin) is connected to the Nano-board **A4**.

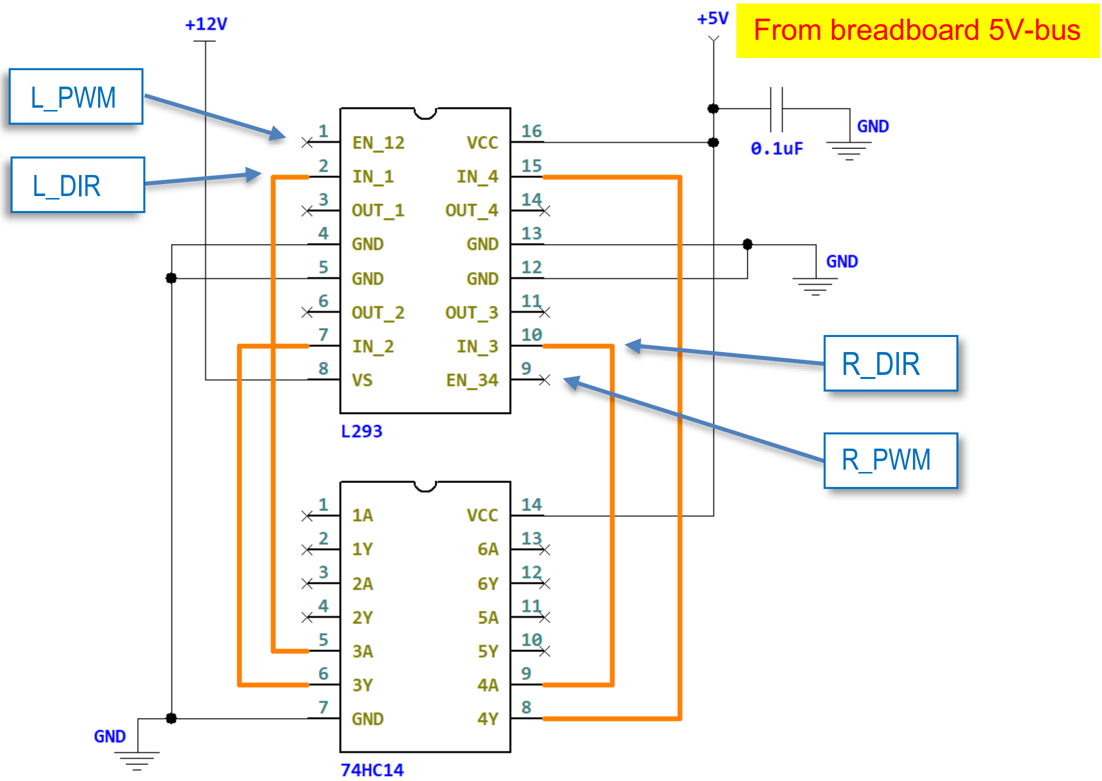
Step 2: Confirm that the **L_Sensor** and **R_Sensor** (sensor "OUT" pin) are connected to the Nano-board **A5** and **A3**, respectively.

You may refer to the circuit diagram on the next page to check the connections between the Nano-board and the L293 motor driver.

Step 3: Confirm that from the **Nano-board D9** and **D10,** the **L_PWM** and **L_DIR** are connected to the **L293 pin 1** and **pin 2**, respectively.

Step 4: Confirm that from the **Nano-board D11** and **D12,** the **R_PWM** and **R_DIR** are connected to the **L293 pin 9** and **pin 10**, respectively.

Step 5: After you are done checking, plug the car's left motor headers into the breadboard rows connected to the L293 pin 3 (+), 4, 5, and 6 (-), and plug the car's right motor headers into the breadboard rows connected to L293 pin 14 (+), 13, 12 and 11(-).



Step 6: **Download** the "**lab_06_logic**" Arduino sketch program from your Canvas lab page.

It is highly recommended to use Arduino IDE 1.8.19 for reliable compiling performance.

The most recent versions may not align well with your laptop due to various compatibility issues.

Step 7: Replace the question marks (**"???"**) with the corresponding **PWM** and **DIR** values required for the robot car to complete the Lab 6 tasks (**Tasks 1–4** on page 4).

You may refer to the **example coding solution** provided in the **tutorial notes**.

Step 8: **Take away the Nano-board from the IC socket** and upload your complete code using the USB cable.

DO NOT do uploading when the Nano board is still in the IC socket.

This may damage the Nano-board because of high current load on pins.

Step 9: Put the Nano-board back into the IC socket after uploading the sketch.

****** TA Check 4: Demo to your TA the successfulness of Tasks 1-4 on the small mat ******

You have learned all the basics, and you are now ready for your project!